

Business Requirements Document (BRD)

Project: Enterprise HR Analytics for Attrition & Engagement

1. Executive Summary

The objective of this project is to build a scalable and analytics-driven HR data platform using Azure Synapse Analytics. The platform will consolidate employee-related datasets from multiple HR systems and generate insights related to employee attrition, engagement, resignation drivers, and early churn signals.

The solution will leverage Azure Data Lake Storage (ADLS Gen2), Azure Data Factory, Azure Synapse Analytics, and Azure Machine Learning to support data engineering, reporting, and predictive analytics use cases.

2. Business Objectives Predict employee attrition risk proactively Identify key resignation drivers across departments and business units Measure employee engagement and employee Net Promoter Score (eNPS) Detect early churn signals using behavioral and survey data Enable HR leadership with data-driven decision making

3. Source Systems

Source System	File Name	Description
Employee System (HRMS)	employee.csv	Employee master and organizational data
Performance System	performance.csv	Employee performance and appraisal metrics
Survey Tool	survey.csv	Employee engagement survey responses and eNPS
Exit Interviews	exit.csv	Employee exit feedback and resignation reasons

4. Data Pipeline Architecture – Medallion Framework

Bronze Layer (Raw Layer) Ingest raw CSV files from external HR systems into ADLS Gen2 Maintain immutable source data for auditing and lineage Capture ingestion metadata and monitoring logs **Silver Layer (Clean Layer)** Clean and standardize employee datasets Perform deduplication and schema validation Apply business rules and transformation logic Create join-ready integrated datasets **Gold Layer (Curated Layer)** Build dimensional star schema model Create aggregated KPI datasets Support reporting, dashboards, and machine learning

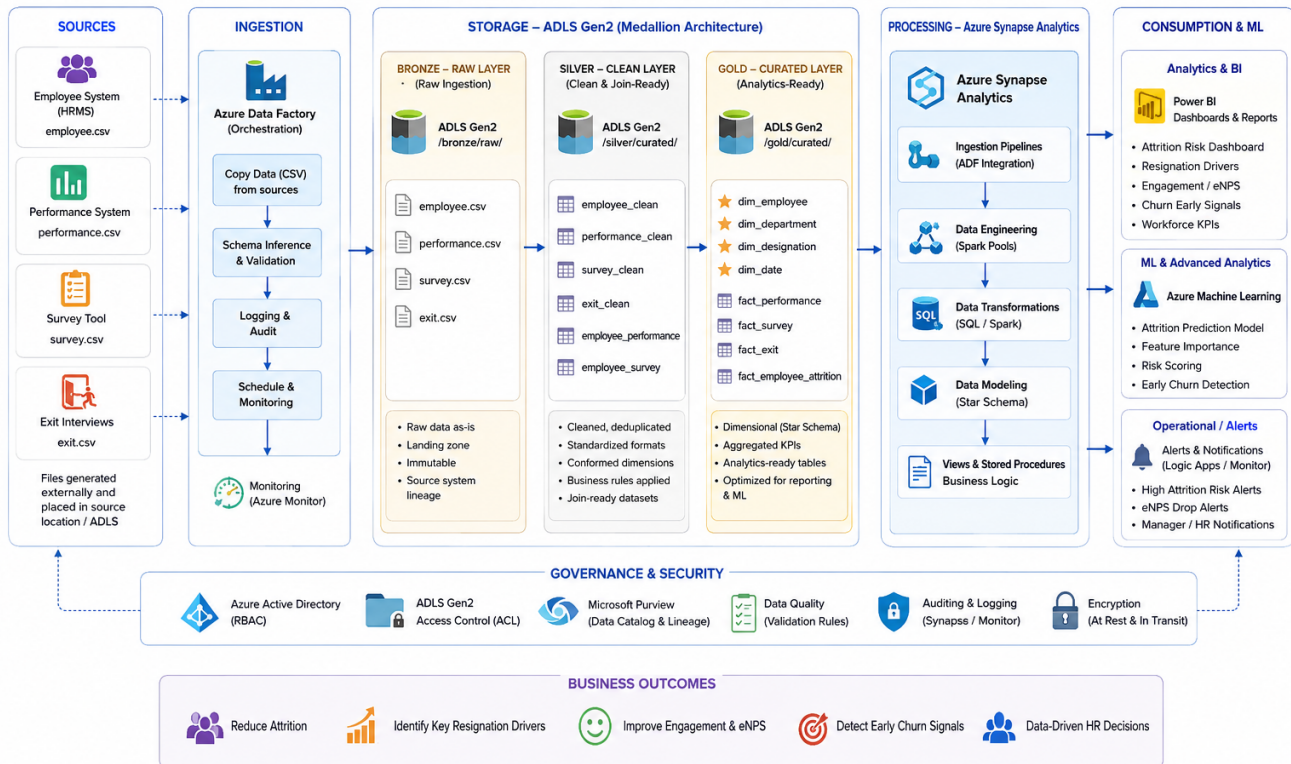
5. Functional Requirements Automated ingestion of CSV files into ADLS Data quality validations and audit logging Transformation pipelines using Azure Synapse Spark/SQL Creation of analytics-ready dimensional models Power BI dashboards for HR insights Attrition prediction and churn detection models Role-based access control and governance

6. Non-Functional Requirements Scalable architecture supporting enterprise HR data growth Secure access using Azure Active Directory and RBAC High availability and monitoring using Azure Monitor Support near real-time analytics refresh cycles Data encryption at rest and in transit

7. End-to-End Architecture Diagram

Enterprise HR Analytics for Attrition & Engagement – End to End Architecture on Azure Synapse

Objective: Predict employee attrition | Identify resignation drivers | Measure engagement & eNPS | Detect early churn signals



8. Expected Business Outcomes Reduction in employee attrition rates Improved employee engagement and satisfaction Early identification of high-risk employees Improved HR operational efficiency Centralized enterprise HR analytics platform

9. Assumptions & Dependencies Source systems will generate CSV files in the agreed format Azure infrastructure and networking are pre-configured Business stakeholders will provide KPI definitions Data governance policies are defined by HR and IT teams